

Participant-Tier Advanced Engine — Implementation Spec

Optimal-Transport drift + nested Active Inference, on-device

Scope. This spec defines the *participant-tier* of the advanced early-warning engine in the Trial-Termination Early-Warning Dashboard: the **on-device**, zero-egress detector that surfaces emerging dose-limiting-toxicity (DLT) and safety risk **earlier and more continuously** than threshold rules, and recommends the most informative next assessment. It is written so a programmer can build it.

Governance (binding). Everything here is **decision-support**. It produces scores and suggestions for the medical monitor / SRC — never a determination. The rule-based triggers (DLT criteria, Hy's-Law/eDISH, QTcF, PK-exposure caps) remain **authoritative**; the validated safety database owns every reported number. The active-inference "action" only ever recommends what to observe (an assessment), for human approval — it never changes dosing, escalation, or termination. The model and its reference barycenters must be **frozen and version-locked** before any use that informs a regulated decision. All participant-level computation runs on the local model with zero egress (Part 11 / ALCOA++ audit).

1. Where it runs and what it touches

Engine	On-device / on-prem local LLM + a small numerical service (NumPy/SciPy-class). No participant-level data leaves the validated environment.
Inputs (per participant, per cohort)	Central-lab safety panel (LFTs: ALT, AST, ALP, total bilirubin; chem; heme), ECG (QTcF), PK exposure (Cmax, AUC vs NOAEL-scaled cap), vitals, AE/CTCAE grades, dosing/visit timing. All already on-device.
Reference	A meta-learned Wasserstein barycenter per analyte, built from prior cohorts/studies of the same compound/class (Section 4). Frozen per analysis.
Outputs	Three continuous scores per participant — W_drift , $F_surprise$, EFE_action — plus a localized "driver" attribution and a drafted, human-routed alert. Written to the dashboard Participant panel + feed with a Part-11 record.
Cadence	Re-score on every new safety/PK posting (near-real-time), and on a fixed daily sweep.

Authoritative-vs-advisory split (must be enforced in code): the rule-based DLT/Hy's-Law/QTcF/exposure checks run **first and independently**; this engine runs alongside and can only *raise attention earlier*. A W_drift /F signal never suppresses or overrides a rule trigger.

2. Signal 1 — Wasserstein drift (Optimal Transport)

Idea. Instead of waiting for a value to cross a line, measure how far a participant's (or cohort's) *distribution* of an analyte has moved from the meta-learned reference, in the **ground metric of clinical severity**.

2.1 Ground metric

Define $d(x, y)$ on each analyte's value space so distance reflects clinical danger, not raw units:

- **LFTs / continuous safety:** work in $\times\text{ULN}$ space (so the metric is comparable across analytes and aligned to the eDISH thresholds). Use $d(x, y) = |g(x) - g(y)|$ with $g = \log_2(\text{value}/\text{ULN})$ so a doubling is one unit and tail moves toward Hy's-Law territory dominate.
- **QTcF:** d in ms (optionally asymmetric — penalize prolongation more than shortening).
- **PK exposure:** d in units of the NOAEL-scaled margin (multiples of the cap).
- **Ordinal CTCAE grade:** $d(i, j) = |i - j|$ (or a convex, grade-escalation-weighted cost).

2.2 Distance

1-D Wasserstein- p has a closed form via the inverse CDF (a sort) — cheap and exact:

```
W_p(μ, ν) = ( ∫_0^1 | F_μ^{-1}(t) - F_ν^{-1}(t) |^p dt )^{1/p}      # 1-D: sort both samples,
integrate quantile gap
```

Use $p = 2$ (penalizes large moves — the dangerous ones — more than W_1). For the **multivariate** participant vector (LFT panel + QTcF + exposure jointly), use **sliced-Wasserstein**, which averages 1-D projections and stays light enough for on-device:

```
SW_2^2(μ, ν) = E_{θ ~ U(S^{d-1})} [ W_2^2( θ·μ, θ·ν ) ] ≈ (1/L) ∑_{l=1..L} W_2^2( proj_{θ_l} μ,
proj_{θ_l} ν )
```

with L fixed projection directions (e.g. $L = 64$), frozen with the model for reproducibility.

2.3 Drift score and localization

$W_drift(\text{participant}) = SW_2(\text{participant_window}, \text{reference_barycenter})$,
computed over a rolling window (e.g. the participant's on-treatment trajectory to date).

Localization (why the flag is actionable): the 1-D optimal coupling on the worst slice is a monotone rearrangement; report the analyte(s) and the value band contributing most to the distance — e.g. *"drift driven by the upper-tail of ALT ($\times\text{ULN}$), day 7→9."* This is what the medical monitor reads.

2.4 Trigger

Maintain a **control band** for W_drift from the reference (Section 4): flag when W_drift leaves the band **and** is trending toward worsening (sign of the tail move). Bands are per-analyte and per-cohort-position (later cohorts expect more drift as dose climbs — encoded in the reference).

```
def w_drift(participant_window, ref_bary, dirs, p=2):
    # participant_window, ref_bary: arrays in ground-metric space (xULN, ms, margin-multiples)
    s = sliced_wasserstein(participant_window, ref_bary, dirs, p)      # SW_2
    driver = worst_slice_attribution(participant_window, ref_bary, dirs) # analyte + value band
    return s, driver

def drift_flag(s, band, trend):
    return (s > band.upper) and (trend > 0)    # advisory only; never suppresses a rule trigger
```

3. Signal 2+3 — Active Inference (free-energy surprise + EFE action)

Generative model (per participant). A small state-space model of the *expected* on-treatment trajectory: latent health state s_t , observations o_t (the analyte vector), transition $p(s_t | s_{t-1})$ and likelihood $p(o_t | s_t)$ whose parameters are the **meta-learned priors** (Section 4). The **preferred observations** C encode the safety/quality tolerances (QTLs, exposure caps, expected analyte ranges).

3.1 Signal 2 — Variational free energy = surprise (early warning)

On each new observation, update beliefs $q(s)$ by minimizing free energy (variational Bayesian belief-updating). The **same F has two equal forms** — one shows the bound, one is what you compute:

```
F = D_KL[ q(s) || p(s | o) ] - ln p(o) ≥ - ln p(o) # the bound (KL ≥ 0): F upper-bounds the surprise
= D_KL[ q(s) || p(s) ] - E_q[ ln p(o | s) ] # complexity - accuracy (what you compute)
```

Under exact inference the bound is tight ($F = -\ln p(o)$ when $q = p(s|o)$); with a restricted family it is a generally non-tight upper bound. Either way, **`F\_surprise` rising = the participant is doing something the model didn't expect** — an anticipatory deviation signal, *before* a frank threshold event. Precision-weight per participant; flag when F exceeds its rolling band.

Watch **two** distinct things: **surprise** (F rising — unexpected) and **preference-divergence** (outcomes drifting from the preferred QTL state, scored against C). They usually move together, but a trajectory can drift toward a fully predicted bad state (low surprise, high preference-divergence) — catch both.

3.2 Signal 3 — Expected Free Energy = what to monitor next

Choose the next monitoring/data-collection action (e.g. an unscheduled LFT/ECG/PK draw) by **minimizing expected free energy** over candidate observations:

```
π* = argmin_π G(π), G(π) = -E[ information gain ] - E_q[ ln p(o | C) ]
                        [ epistemic ]                [ pragmatic (utility) ]
```

The agent **minimizes G**; because both terms are negated, that **maximizes** expected information gain (the epistemic term — pick the assessment that most reduces uncertainty about emerging DLT risk) and the expected log-preference $\ln p(o|C)$ (the pragmatic term — favor reaching the safe, on-track state; C is the utility). Output: the highest-value assessment, surfaced to the medical monitor **as a suggestion to approve**, with its expected information gain shown.

```
def free_energy(q, prior_s, loglik): # F = complexity - accuracy
    return kl(q, prior_s) - expect(q, loglik)

def efe(candidate_obs, q, model, C): # rank candidate assessments
    scored = []
    for a in candidate_obs: # e.g. {LFT, ECG, PK draw}
        epistemic = expected_info_gain(a, q, model) # mutual information
        pragmatic = expected_log_pref(a, q, model, C) # E_q[ ln p(o|C) ]
        G = -(epistemic) - (pragmatic)
        scored.append((a, G, epistemic))
    a_star, G_star, ig = min(scored, key=lambda r: r[1]) # minimize G
    return Recommendation(action=a_star, value_of_info=ig) # human approves; never auto-ordered
```

4. The meta-learned reference (nesting + learning-to-learn)

The participant tier is the **bottom of a deep active-inference hierarchy** (participant → study → client). Its priors are **supplied from above** and **meta-learned across studies**:

- **Reference barycenter.** Per analyte, $\mu_{\text{ref}} = \text{argmin}_{\mu} \sum_k \lambda_k W_2^2(\mu, v_k)$ over prior cohorts v_k (Wasserstein-2 barycenter). This is the distribution a new participant/cohort is scored against. The **control bands** in §2.4 / §3 are derived from the spread of $\{v_k\}$ around μ_{ref} .
- **Generative-model priors.** Transition/likelihood parameters and the preferred set C are empirical-Bayes estimates pooled across prior cohorts/studies (a **hyperprior**). The higher tier *supplies* the prior to the participant tier; what is *meta-learned* from the population is the hyperprior — so a new cohort starts **population-informed**, not from scratch, and recalibrates as each study completes.
- **Coupling up.** A participant W_{drift}/F excursion is a bottom-up prediction error that propagates to the study tier (raising cohort-level risk) — the nesting, not three separate dashboards.
- **Optional unifier.** A *Wasserstein-regularized objective* (OT cost in place of/alongside the KL term) makes the belief-updating geometry-aware and lets the barycenter prior enter directly. **Caveat:** once $KL \rightarrow W$ it is no

longer the strict variational free energy, so the "upper bound on surprisal" guarantee relaxes — a deliberate, research-grade option, off by default.

```
def fit_reference(prior_cohorts):          # one-time, frozen per analysis
    bary = wasserstein_barycenter(prior_cohorts, p=2)  #  $\mu_{\text{ref}}$  per analyte
    bands = control_bands(prior_cohorts, bary)        #  $W_{\text{drift}}$  / F bands
    prior = empirical_bayes_hyperprior(prior_cohorts) # transition/likelihood + C
    return frozen(Reference(bary, bands, prior))      # version-locked artifact
```

5. Roll-up, output, and the alert

For each participant: `participant_score = w(  $W_{\text{drift}}$ ,  $F_{\text{surprise}}$ ,  $\text{pref\_divergence}$  )` with **fixed, documented weights** (not a learned black box) → maps to the Participant-panel RAG. Any amber/red:

1. localizes the driver (analyte + value band + day),
2. pulls the rule-based context (current DLT tally vs the design's stopping rule, eDISH position, QTcF, exposure vs cap) **from the validated tools**,
3. drafts the alert and routes it to the **medical monitor** / **SRC chair** *ahead* of the scheduled cohort review, with the EFE-recommended next assessment attached,
4. writes a Part-11 record; a human dispositions it. Nothing is paused/escalated by the system.

6. Validation plan (before any regulated use)

Step	What	Acceptance
Freeze	Lock model weights, the L projection directions, μ_{ref} , control bands, roll-up weights as a versioned artifact.	Re-run on the same inputs is bit-stable.
Back-test / lead-time	Replay completed cohorts; measure how many days <i>earlier</i> W_{drift} /F flagged a confirmed DLT/Hy's-Law vs the rule trigger.	Median lead-time > 0; documented per analyte.
Operating characteristics	On historical data, estimate sensitivity / false-alert rate / alerts-per-cohort-week at the chosen bands.	Alert burden acceptable to the medical monitor; sensitivity $\geq$ agreed floor.
Concordance	Confirm the engine never <i>suppresses</i> a rule trigger; agreement of "driver" attribution with adjudicated cause.	100% rule-trigger pass-through; attribution reviewed.
Human-factors	Medical monitor reviews drafted alerts + EFE suggestions for usefulness and non-coercion.	Suggestions read as advisory; no auto-action path exists.
Documentation	Spec, validation report, frozen-artifact hash, Part-11 lineage.	Submission-grade package.

7. Build order

1. **Ground metrics + 1-D / sliced Wasserstein** on one analyte family (LFTs), reference barycenter from prior cohorts, W_{drift} + driver attribution, control bands. (*Cheap, high-value, fully on-device.*)
2. **Free-energy detector** (F_{surprise}) on the same participant state-space model; add preference-divergence vs C.

3. **EFE recommender** (π^*) over the protocol-allowed assessment set; wire the human-approval queue.
4. **Roll-up + alert drafting + Part-11 record + dashboard write.** Extend metrics to QTcF and PK exposure.
5. **Freeze + validation (Section 6).** Only then does any output inform a regulated decision.

Companion to the Trial-Termination Early-Warning Dashboard (`Trial\_Risk\_EarlyWarning\_Dashboard.html`, page  Advanced engine”). Decision-support; advanced/research-grade; the rules and the humans remain authoritative.